Deterministic Systems Lenses

$$\begin{pmatrix} S \\ S \end{pmatrix} \underset{\text{readout}}{\overset{\text{update}}{\rightleftarrows}} \begin{pmatrix} I \\ O \end{pmatrix} \rightleftarrows \begin{pmatrix} I' \\ O' \end{pmatrix}$$

update : $S \times I \to S$

readout : $S \to O$

General (Deterministic) Lens Type Signatures

$$X \xleftarrow{f^\#} A \times Y \qquad Y \xleftarrow{g^\#} B \times Z$$

$$\begin{tikzcd}
\begin{pmatrix} X \\ A \end{pmatrix} \arrow[r, shift right, "f"'] & \begin{pmatrix} Y \\ B \end{pmatrix} \arrow[l, shift right, "f^\#"'] \arrow[r, shift right, "g"'] & \begin{pmatrix} Z \\ C \end{pmatrix} \arrow[l, shift right, "g^\#"']
\end{tikzcd}$$

$$A \xrightarrow{f} B \xrightarrow{g} C$$

$$X \xleftarrow{f^\#} A \times Y \qquad Y \xleftarrow{g^\#} B \times Z \qquad Z$$

Cokleisli Maps of the Product Comonad!

Product Comonad $C \times -$

$$\begin{tikzcd} C \times C \times X & \\ C \times X \arrow[u, "\kappa"'] \arrow[d, "\varepsilon"] & \\ X & \end{tikzcd}$$

CoKleisli Category:

Morphisms have signature

$$C \times X \longrightarrow Y$$

Exercises:

TEDIOUS, BUT LIKELY USEFUL

Monoidal categories:
* Fully understand monoidal categories, and their structures such as strict, lax, strong, weak, or whatever.
    Also consider the coherence diagrams

Comonads:
* Fully define a comonad, starting with the phrase "A comonad is a comonoid in the category of endofunctors."
    State all definitions in full depth, and maybe use string diagrams to help. Make note of which structures are available and where they are required. Ask, for instance, is the category of endofunctors strict or lax or strong or weak or whatever? How do those structure morphisms interact with the definition of a comonad?

* Fully define a coKleisli category, and prove that it's a category (prove that it's associative and unital). Make note of which structures stated above are required where.

* Define the $C\times\dash$ comonad, and check that it satisfies all conditions noted above.

* Define $CoKl(C\times\dash)$. Walk through the composition computation and make note of the previously identified structures and where they're used

ACTUALLY INTERESTING
* Consider a general monoidal category, be it lax, weak, strong, strict, or whatever. What minimal structures do we need to make $C\otimes\dash$ into a monad? Is a Markov category sufficient?

* What about definiting something like $CoKl(C\otimes\dash)$? Do we even need a monad? As we remove more and more structures, how does that affect the resulting coKleisli category?

* What about adding structures, such as symmetry, and compatibility between the monad and monoidal structure?

Problem:

The left objects are different!

$$\overbrace{B \times Z \qquad\qquad A \times Y}$$

$$B \times Z \xrightarrow{g^\#} Y \qquad A \times Y \xrightarrow{f^\#} X$$

$$Z$$

It's easy to do composition within $\mathrm{CoKl}(C \times -)$

But $g^\# \in \mathrm{CoKl}(B \times -)$ and $f^\# \in \mathrm{CoKl}(A \times -)$

ie. $\mathrm{CoKl}(C \times -)$ is indexed over varying C

Indexed Categories

Defn: given a category $\mathcal{C}$, an indexed category is a functor

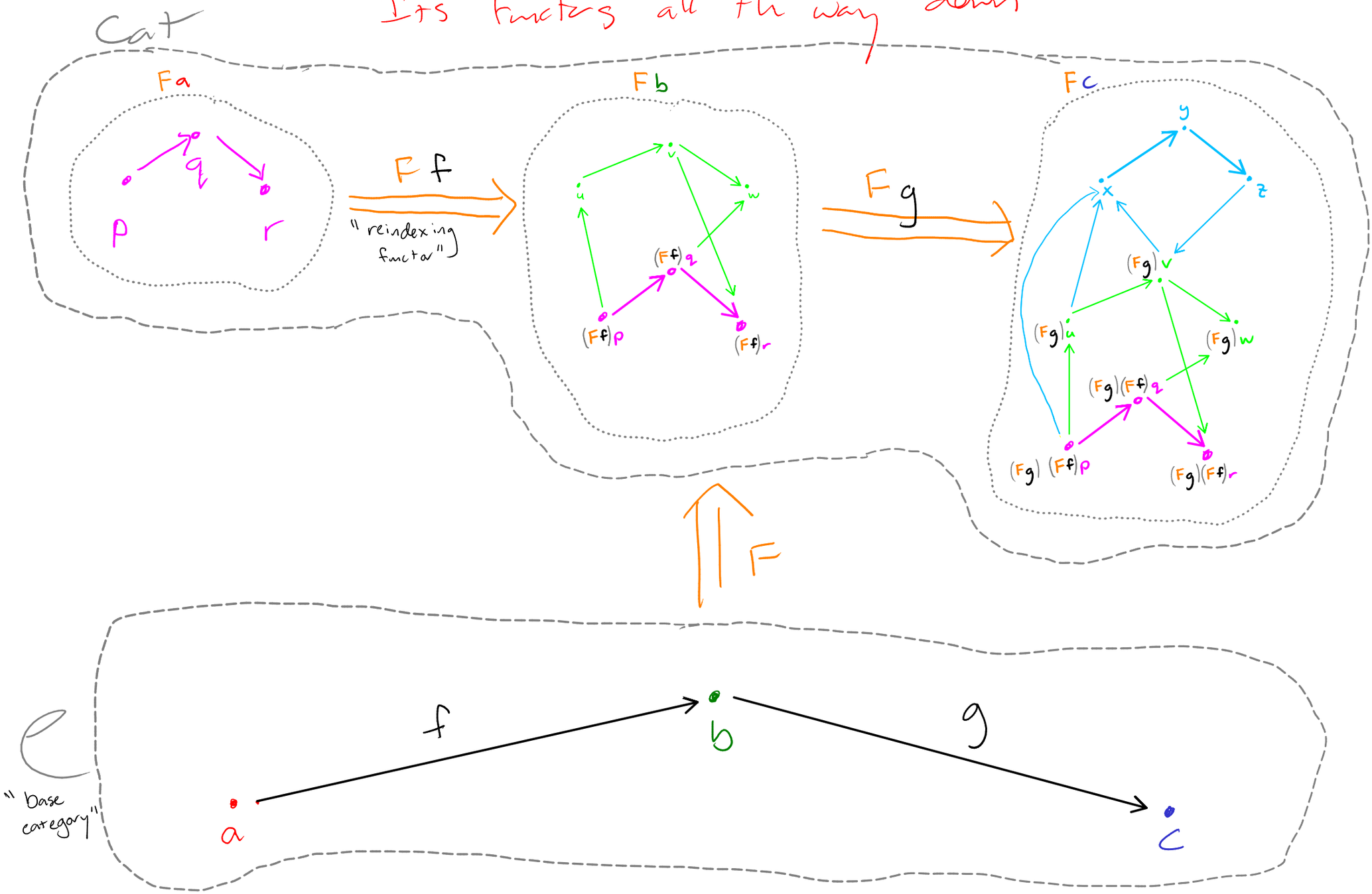
$$F : \mathcal{C} \longrightarrow \mathrm{Cat} \quad \text{(covariant)}$$

or

$$F : \mathcal{C}^{op} \longrightarrow \mathrm{Cat} \quad \text{(contravariant)}$$

Anatomy of an indexed cat

It's functors all the way down



Exercises

* Define the indexing functors for $CoKl(C\times\dash)$. Make sure it satisfies everything for an indexed category, such as functoriality, ma
  functor compositions are associative, etc.

* What happens if structure is removed or added? What if the functors become lax, or weak, or whatever?

* Can you transport any monoidal stucture from the base category to the indexed category? How? What if the indexed category already ha

* If the base category C is Markov, does our hypothetical definition of $CoKl(C\otimes\dash)$ make an indexed category?
  If not, what about over $\mathsf{C}_{\mathsf{det}}$? In other words, can deterministic morphisms generate reindexing functors if g

* If the base category is Cartesian, supposedly $CoKl(\times\dash)$ is not. Explore why.

How to compose morphisms in $F(X)$ with those in $F(Y)$? Ans: The Grothendieck Construction

The notation is $\int_{C \in \mathcal{C}} F(C)$, because it's kind of like an integral.

Defn: Given $F : \mathcal{C} \longrightarrow \mathrm{Cat}$, $\int_{C \in \mathcal{C}} F(C)$ has

(covariant)

Objects

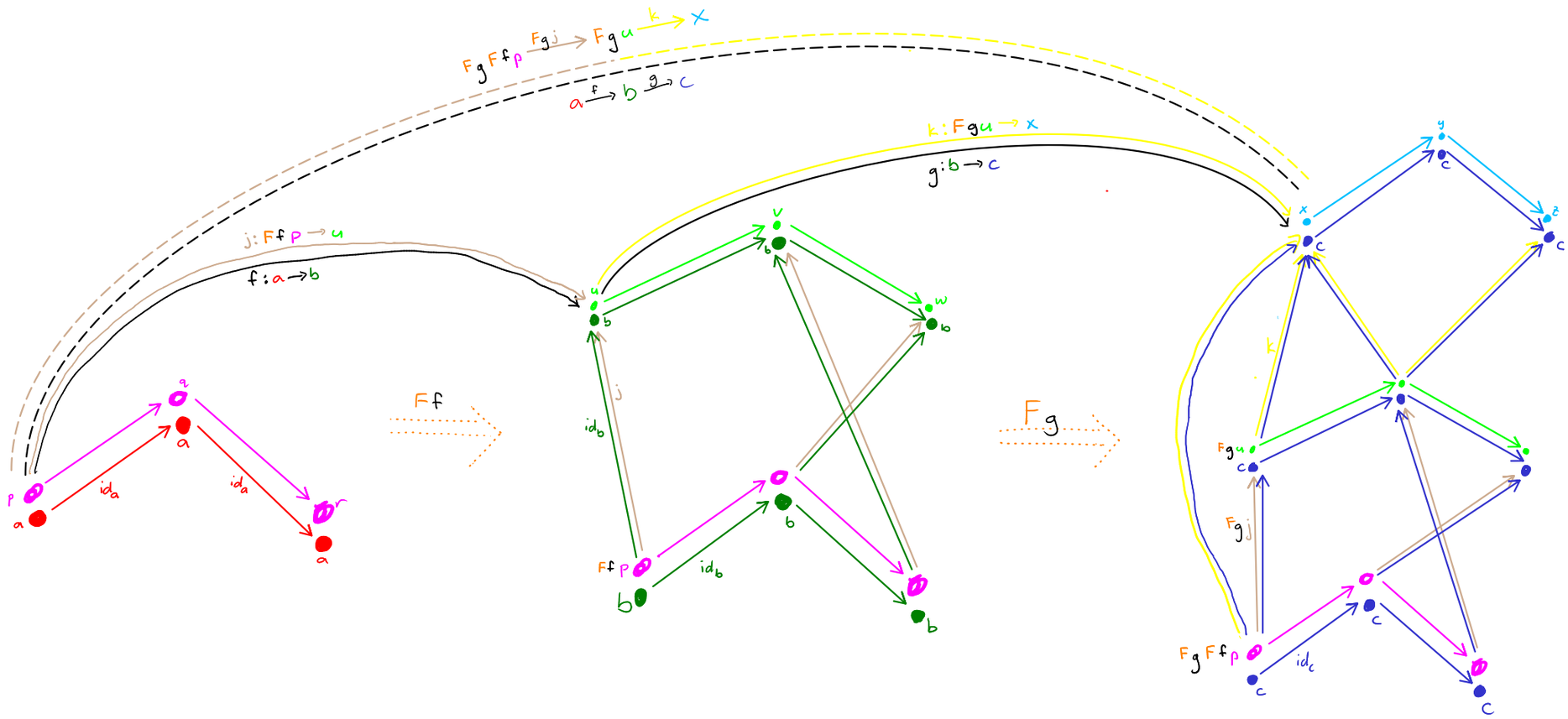
all pairs (A, X)

such that $A \in \mathcal{C}$

$X \in F(A)$

Morphisms $(A, X) \longrightarrow (B, Y)$

are pairs

$$\Big(\underbrace{A \xrightarrow{f} B}_{\in \mathcal{C}} \, , \, \underbrace{(Ff)[X] \xrightarrow{g} Y}_{\in F(B)} \Big)$$

Anatomy of the (covariant) Grothendieck Construction